

The Revealing Guide to Smartphone Battery Survival

Summary

The history of batteries can be traced back over two thousand years to ancient Iraq. The Baghdad battery, an unglazed terracotta jar discovered in the capital city, was a device that use copper and iron to function. Nowadays, highly developed lithium batteries have entered all aspects of our lives, especially in the construction of modern smart phones.

The point is, the heavier we are going to rely on it, the more imperative a comprehensive understanding becomes. In this paper, we develop a comprehensive mathematical physics model, the **Continuous-Time Model (CTM)**, to demonstrate the **state of charge (SOC) dynamics** and **state of health (SOH) progression** under realistic operating conditions.

First, we propose **three** models. We establish the **Battery Electrical Model (BEM)** based on an **Internal Resistance (R_{int}) circuit**. We derive an explicit analytical expression for discharge current, ensuring computational efficiency and physical intelligibility. We propose the **Proactive Consumption Model (PRCM)** to decompose system-level power demand. We map major subsystems—including heterogeneous CPU clusters, 5G/Wi-Fi communication, and display modules—into duty-cycle and utilization-based power components. This allows us to translate complex user behaviors into quantified electrical loads. To account for long-term capacity fade, the **Passive Consumption Model (PACM)** is established. This model incorporates **Arrhenius-driven calendar aging** and cycle aging. Notably, we introduce a **Lithium Plating Punishment** factor to simulate accelerated degradation during low-temperature charging, providing a robust tool for SOH prediction.

Next, we formally present the **Continuous-Time Model (CTM)**, by integrating the models above. Using a real-time data set, we initially adopt a **static approach** to validate our model. Unsatisfied with the fit of 0.96, we then develop a **dynamic approach**, which resulted in a better fit of 0.97, perfectly matching the target curve's shape, as well. Closely after, we use the CTM to calculate time-to-empty in terms of SOH, also using both the static and dynamic methods. Among all predicted data points, we selected a suitable portion to **statistically analyze its major drivers**.

What's more, we further explore the battery's state of health (SOH) by making full use of our **Passive Consumption Model (PACM)**, where we also take **environmental temperature** into consideration, which is caused by **season and geographic location**.

Finally, we perform sensitivity analysis on some of the key parameters. Recommendation are provided for both users and developers, based on our findings. We end our paper with an analysis of strengths and limitation.

Keywords: Smart Phone Battery, Power Sum Framework, Continuous-Time Model of SOC, Time-to-Empty, SOH, Battery Aging, Arrhenius factor, Lithium Plating

Contents

1	Introduction	2
1.1	Problem Background	2
1.2	Restatement of the Problem	2
1.3	Our Work	3
2	Assumptions and Justifications	3
3	Notations	4
4	Models Proposal	5
4.1	Battery Electrical Model (BEM)	5
4.2	Proactive Consumption Model (PRCM)	7
4.3	Passive Consumption Model (PACM)	10
5	Continuous-Time Model of SOC (CTM)	11
5.1	Fully Closed SOC Model	12
5.2	Data-driven Validation	13
6	Time-to-empty Prediction	15
6.1	Model Overview	15
6.2	Model Details	15
6.3	Model Evaluation	16
7	Extension on State of Health (SOH)	17
8	Sensitivity Analysis	19
8.1	Proactive Discharge Dynamics	19
8.2	Aging Correction	20
9	Recommendation	20
9.1	For Smartphone Users	20
9.2	For Operating System (OS) Designers	21
10	Strengths and Limitations	21
10.1	Strengths	21
10.2	Limitations	21
11	Conclusion	22
	References	23
	Report on Use of AI Tools	24

1 Introduction

1.1 Problem Background

Smartphone battery performance often exhibits significant variability in daily use, making it difficult to explain using simple user intuition. Even for the same device, battery life may differ substantially from day to day, indicating that energy depletion is not governed by a single dominant factor but rather arises from the combined effects of multiple factors.

From a system-level perspective, energy consumption results from the coordinated operation of multiple components, including computational workloads, display modules, wireless communication, and various functional modules scheduled by the operating system. These processes exhibit pronounced temporal dynamics and frequently occur simultaneously, causing system power demand to evolve continuously rather than remaining constant or piecewise static. Consequently, even when total usage time is similar, short periods of intensive activity, prolonged moderate workloads, and the cumulative effects of background tasks can lead to markedly different discharge behaviors.

In addition to system activity, external environmental conditions and historical factors also exert significant influence on battery behavior. Variations in ambient temperature can alter the internal characteristics of the battery, while long-term usage patterns and charging habits gradually affect its effective capacity and response to load. These effects accumulate over time and interact with instantaneous power demand, making battery dynamics difficult to characterize using simple or purely empirical approaches.

Therefore, a deeper understanding of smartphone battery drain requires a modeling framework that can describe time-dependent power consumption, reflect the physical behavior of the battery, and maintain strong interpretability. By linking system-level power demand with battery-side electrical behavior on a unified time scale, such a model provides a necessary foundation for characterizing the evolution of the state of charge and for making reliable predictions of the remaining time to empty under realistic operating conditions.

1.2 Restatement of the Problem

Considering the background information and constraints described in the problem statement, the following problems are addressed in this study:

- Develop a **continuous-time governing differential equation** (or system of equations) to describe the evolution of the **state of charge (SOC)** of a lithium-ion smartphone battery under realistic usage conditions.
- Extend the SOC dynamics to account for **battery aging effects** and **long-term degradation**, and assess how capacity fade and the associated evolution of the **state of health (SOH)** influence SOC evolution and prediction accuracy.
- Build a structured model to decompose **system-level power consumption** into its major contributing components and map the resulting power demand to battery-side electrical variables required by the governing equations.

- Create a method to predict the **remaining time-to-empty (TTE)** for different initial SOC levels and usage scenarios, and identify the dominant drivers of rapid battery drain.
- Develop a sensitivity and uncertainty analysis framework to evaluate how modeling assumptions, parameter variations, and fluctuations in usage patterns affect **SOC, SOH, and TTE** predictions.
- Design a strategy to translate modeling insights into practical recommendations for users and operating systems aimed at improving battery life.

1.3 Our Work

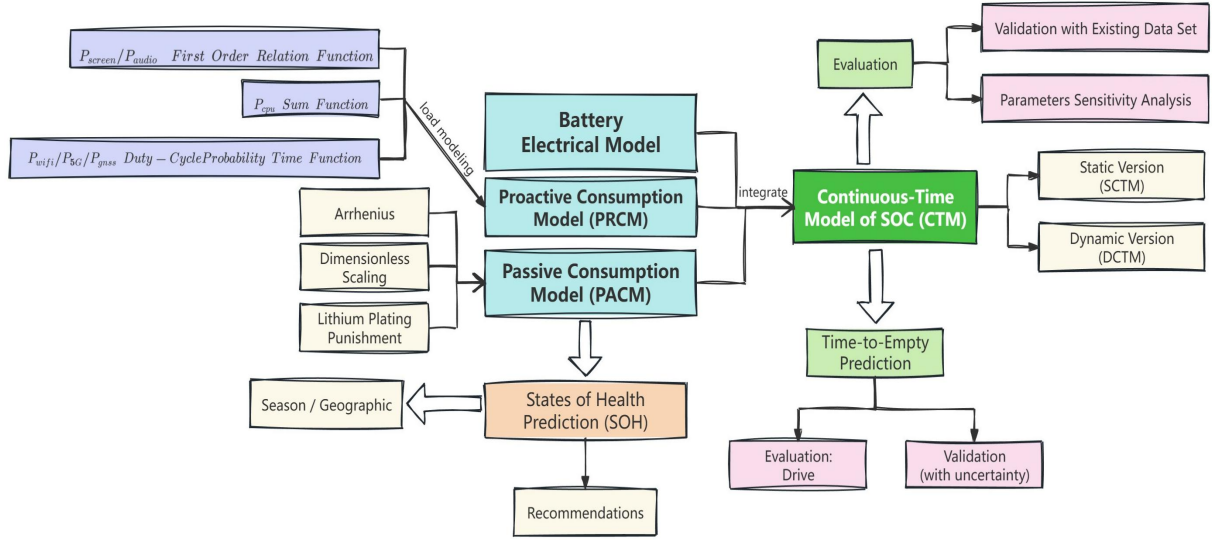


Figure 1: Flowchart of our work.

2 Assumptions and Justifications

To simplify the problem, we make the following basic assumptions, each of which is properly justified.

Assumption 2.1. The electrical behavior of the smartphone lithium-ion battery is represented by an **Internal Resistance (R_{int}) circuit model** consisting of an open-circuit voltage source and an equivalent series resistance.

Justification: At the time scale relevant to SOC and TTE analysis, higher-order polarization effects introduce only secondary influences and are therefore neglected.

Assumption 2.2. The battery charging and discharging processes are continuous in time, and SOC is treated as a continuous and differentiable state variable.

Justification: Under typical operating conditions, the battery charging and discharging processes exhibit gradual temporal variation, supporting the formulation of SOC dynamics using continuous-time differential equations.

Assumption 2.3. The total system power consumption is decomposable into an **additive combination** of subsystem power components and remains approximately constant over short time intervals.

Justification: Within the numerical integration window, power fluctuations exert limited influence on SOC evolution and can be approximated as constant for tractability.

Assumption 2.4. The internal resistance of the battery is assumed to be constant within typical operating SOC and temperature ranges.

Justification: Compared with variations in system power demand, resistance changes introduce only secondary effects on SOC and TTE prediction.

Assumption 2.5. Battery aging is assumed to manifest primarily as **effective capacity degradation** at the system level.

Justification: Capacity fade provides a practical and interpretable representation of dominant long-term aging effects without resolving microscopic mechanisms.

Assumption 2.6. Variations in usage behavior, environmental conditions, and model parameters are bounded and statistically representable.

Justification: This assumption ensures that variations in parameters and usage behavior remain within a tractable range, allowing systematic sensitivity and uncertainty analyses.

3 Notations

Table 1: List of symbols, their definitions, and units

Symbol	Definition	Unit
t	Time	h
SOC	State of charge of the battery	–
SOH	State of health of the battery	–
Q	Available electric charge in the battery	Ah
C	Nominal battery capacity	Ah
I	Battery current (positive during discharge)	A
V	Battery voltage (terminal or open-circuit)	V
R	Battery internal resistance	Ω
P	Battery-side power demand	W
η	Efficiency factor	–
k_{sd}	Self-discharge rate of the battery	h^{-1}
b	Coefficients of the OCV–SOC polynomial fit	V
y	Aging-induced capacity loss rate	$Ah \cdot h^{-1}$
k_{base}	Baseline degradation coefficient in aging model	h^{-1}
k_{cal}	Calendar aging sensitivity factor	–
n	Cycle aging current exponent	–
T	Battery temperature	K
E_a	Activation energy of degradation reactions	$J \cdot mol^{-1}$
R_g	Universal gas constant	$J \cdot mol^{-1} K^{-1}$
TTE	Time to empty	h
σ	Uncertainty or standard deviation	–

4 Models Proposal

4.1 Battery Electrical Model (BEM)

This model is proposed to establish the basic structure of the circuit contained inside a modern smart phone.

Sign convention The battery current $I(t)$ is defined to be positive during discharge, i.e., when electrical energy is delivered from the battery to the smartphone load. Accordingly, the system-level power demand satisfies $P_{\text{total}}(t) \geq 0$ under normal usage.

Internal Resistance (R_{int}) model At the time scale relevant for state-of-charge (SOC) evolution, the battery is modeled using a standard **Internal Resistance model (R_{int} model)**. This equivalent circuit consists of an open-circuit voltage source $V_{\text{OC}}(\text{SOC})$ in series with an internal resistance $R_{\text{int}}(\text{SOC}, T)$. The terminal voltage is therefore given by

$$V_{\text{cell}}(\text{SOC}(t), T(t)) = V_{\text{OC}}(\text{SOC}(t)) - I(t) R_{\text{int}}(\text{SOC}(t), T(t)). \quad (1)$$

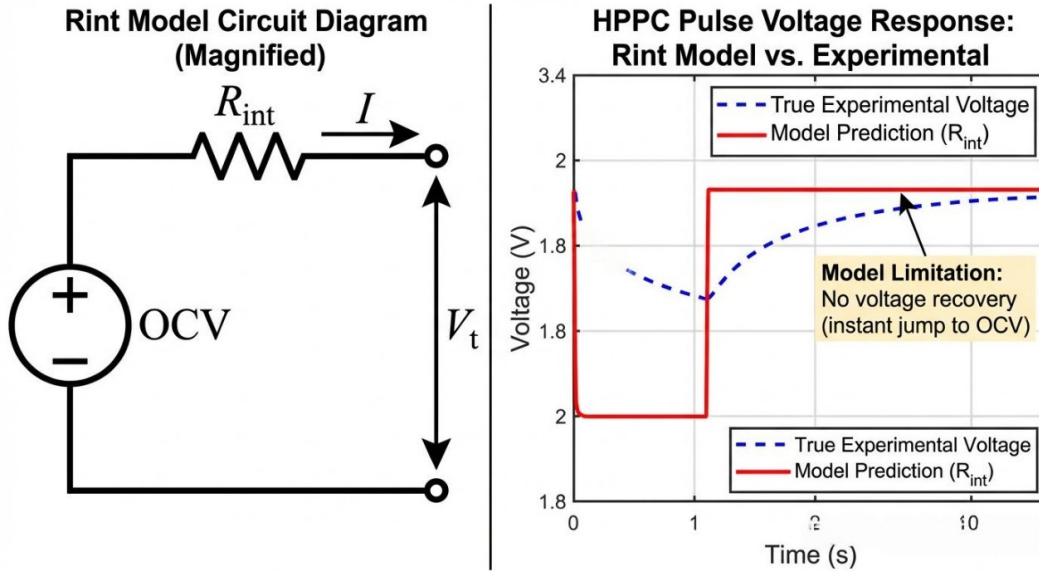


Figure 2: Circuit diagram of the Rint model and comparison of voltage responses.

Power balance relation The smartphone imposes an instantaneous electrical power demand at the battery terminals, which must satisfy

$$P_{\text{total}}(t) = V_{\text{cell}}(\text{SOC}(t), T(t)) I(t). \quad (2)$$

Substituting the terminal-voltage relation into the power balance equation yields

$$P_{\text{total}}(t) = \left(V_{\text{OC}}(\text{SOC}(t)) - I(t) R_{\text{int}}(\text{SOC}(t), T(t)) \right) I(t). \quad (3)$$

Rearranging terms leads to a quadratic equation in the current $I(t)$:

$$R_{\text{int}}(\text{SOC}(t), T(t)) I^2(t) - V_{\text{OC}}(\text{SOC}(t)) I(t) + P_{\text{total}}(t) = 0. \quad (4)$$

4.1.1 Second-Order OCV Approximation

To obtain a closed-form mapping from SOC to voltage, the open-circuit voltage is approximated by a second-order polynomial in SOC,

$$V_{\text{OC}}(\text{SOC}) = b_0 + b_1 \text{SOC} + b_2 \text{SOC}^2, \quad (5)$$

where the coefficients (b_0, b_1, b_2) are obtained by fitting voltage–SOC data using voltage-controlled battery models. Such low-order polynomial representations are widely used for SOC and SOH diagnosis and provide a good balance between accuracy and analytical tractability [1].

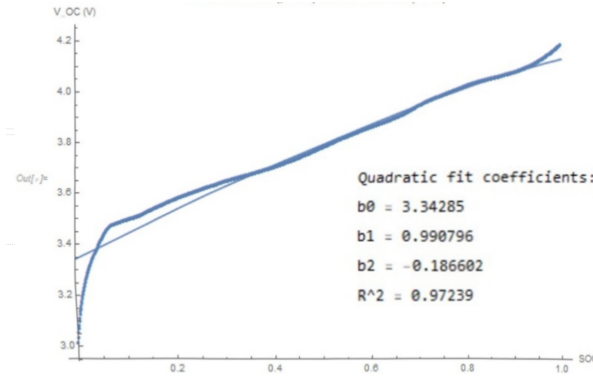


Figure 3: Second-order polynomial fit of the open-circuit voltage as a function of SOC.

4.1.2 Explicit Current Expression

Solving the quadratic equation (4) using the quadratic formula gives

$$I(t) = \frac{V_{\text{OC}}(\text{SOC}(t)) \pm \sqrt{V_{\text{OC}}^2(\text{SOC}(t)) - 4 R_{\text{int}}(\text{SOC}(t), T(t)) P_{\text{total}}(t)}}{2 R_{\text{int}}(\text{SOC}(t), T(t))}. \quad (6)$$

For discharge operation with $P_{\text{total}}(t) > 0$ and $R_{\text{int}} > 0$, the physically meaningful solution satisfies $0 < I(t) < V_{\text{OC}}/R_{\text{int}}$ and corresponds to the negative branch of the square root. The

resulting **explicit current expression** is therefore

$$I(t) = \frac{V_{OC}(SOC(t)) - \sqrt{V_{OC}^2(SOC(t)) - 4 R_{int}(SOC(t), T(t)) P_{total}(t)}}{2 R_{int}(SOC(t), T(t))}. \quad (7)$$

4.2 Proactive Consumption Model (PRCM)

Smartphone usage is inherently proactive, so we define the energy associated with normal operation as proactive consumption. The **Proactive Consumption Model (PRCM)** computes the total system power demand.

To relate system-level power demand to battery discharge, we introduce an overall power-conversion efficiency parameter $\eta \in (0, 1]$, which accounts for losses in the PMIC and DC–DC conversion stages. We distinguish between the system-side power consumption $P_{sys}(t)$ and the effective battery-side power demand $P_{total}(t)$. The latter, which enters the battery and SOC models, is given by

$$P_{total}(t) = \frac{1}{\eta} P_{sys}(t). \quad (8)$$

The system-side power consumption is decomposed into a baseline term and incremental contributions from major subsystems:

$$P_{sys}(t) = P_0 + \Delta P_{CPU}(t) + \Delta P_{WiFi}(t) + \Delta P_{GNSS}(t) + \Delta P_{Screen}(t) + \Delta P_{5G}(t) + \Delta P_{Audio}(t). \quad (9)$$

The baseline power P_0 represents the near-idle power floor (screen off, no explicit interaction) and aggregates essential idle costs:

$$P_0 = \sum_{k \in K} P_{k, idle} + P_{SL} + P_{idle} + P_{static}. \quad (10)$$

Here, $k \in K$ indexes CPU clusters and $P_{k, idle}$ is the powered-idle cost of cluster k . P_{SL} denotes wireless standby power, P_{idle} captures connectivity-related idle costs, and P_{static} collects device-level static overheads (e.g., memory retention and always-on services).

Even in apparent idle states, baseline power persists due to OS activity and periodic wakelocks. A partial CPU wakelock draws about 0.09 W [2]. Since true standby does not require continuous wakelocks, we conservatively set

$$P_0 = 0.1 \text{ W}. \quad (11)$$

CPU Power

Modern SoCs use heterogeneous multi-core CPUs organized into clusters with different operating ranges. We model CPU power at the cluster level.

For each CPU cluster $k \in K$, the activity-dependent component is approximated using a

first-order utilization model:

$$\Delta P_{\text{CPU}}(t) = \sum_{k \in K} \alpha_k u_k(t), \quad (12)$$

where $u_k(t) \in [0, 1]$ is the utilization of cluster k , and α_k is the marginal power from idle to full load.

We adopt a three-cluster configuration,

$$K = \{L, M, B\},$$

corresponding to the efficiency, middle, and performance clusters. For two-cluster platforms, the model reduces to $K = \{L, B\}$.

Wi-Fi Power

At the system level, Wi-Fi power is mainly determined by the active communication duty cycle. We use a simplified duty-cycle model:

$$\Delta P_{\text{Wi-Fi}}(t) = D_{\text{Wi-Fi}}(t) (P_{\text{Wi-Fi,ac}} - P_{\text{Wi-Fi,id}}), \quad (13)$$

where $D_{\text{Wi-Fi}}(t) \in [0, 1]$ is the long-term average active duty cycle and $P_{\text{Wi-Fi,id}}$ is absorbed into P_0 .

The active duty cycle is approximated via the long-term average throughput $\bar{R}(t)$:

$$D_{\text{Wi-Fi}}(t) = \min\left(1, \frac{\bar{R}(t)}{R_{\text{eff}}}\right), \quad (14)$$

where R_{eff} is an effective rate accounting for protocol and MAC overhead.

GNSS Power

GNSS operates intermittently; deep-sleep consumption is absorbed into P_0 , and only the incremental active cost is modeled.

Consider a time window $[t, t + \Delta]$. Let $t_{\text{ac}}(t)$ be the total time within this window during which GNSS is active. The GNSS activity duty cycle is defined as

$$D_{\text{GNSS}}(t) = \frac{t_{\text{ac}}(t)}{\Delta} \in [0, 1]. \quad (15)$$

Assuming the GNSS active-state power is approximately constant and denoted by $P_{\text{GNSS,ac}}$, the time-averaged incremental GNSS power over the window is

$$\Delta P_{\text{GNSS}}(t) \approx D_{\text{GNSS}}(t) P_{\text{GNSS,ac}}. \quad (16)$$

To capture weak-signal overhead without explicit environment modeling, we apply a multiplicative perturbation:

$$P_{\text{GNSS,ac}} \mapsto (1 + \delta) P_{\text{GNSS,ac}}, \quad \delta \in [0, 0.5]. \quad (17)$$

Therefore,

$$\Delta P_{\text{GNSS}}(t) = D_{\text{GNSS}}(t) (1 + \delta) P_{\text{GNSS,ac}}. \quad (18)$$

Screen Power

Display power is dominated by refresh-dependent driving losses and brightness-dependent luminous output. We use a first-order incremental model.

Let $f(t)$ denote the refresh rate (Hz), $B(t) \in [0, 1]$ the normalized user brightness level, and $APL(t) \in [0, 1]$ the average picture level (average pixel intensity). Since static display losses are absorbed into the baseline power P_0 , we model the incremental screen power as

$$\Delta P_{\text{Screen}}(t) = \gamma_{\text{scr}} f(t) + \alpha_{\text{scr}} B(t) APL(t), \quad (19)$$

where γ_{scr} and α_{scr} are device-dependent coefficients. For LCD screens, one may approximate $APL(t) \approx 1$.

5G Power

We model only the incremental 5G power above P_0 . In the active radio state, the cost includes an active baseline, a throughput-dependent term, and a signal-quality penalty.

Let $D_{5G}(t) \in [0, 1]$ be the fraction of time the modem stays in an active radio state within a short averaging window. The incremental 5G power is modeled as

$$\Delta P_{5G}(t) = D_{5G}(t) \left(P_{5G,\text{base}} + \beta_{5G} R(t) + \gamma_{5G} \phi(S(t)) \right), \quad (20)$$

where $P_{5G,\text{base}}$ is the active baseline power of the 5G subsystem, $R(t)$ is the data rate (Mbps), and $\phi(S) \geq 0$ is a monotone penalty for poor signal quality.

To avoid unit inconsistency and sign ambiguity, we map the raw signal indicator $S(t)$ (e.g., RSSI/RSRP in dBm) into a nonnegative penalty:

$$\phi(S(t)) = \max(0, S_{\text{ref}} - S(t)), \quad (21)$$

where S_{ref} is a reference signal level.

Audio Power

Audio power is mainly determined by whether the audio processing chain is active; when no audio is played, the incremental audio cost is negligible.

Let $a(t) \in [0, 1]$ denote the audio activity indicator, where $a(t) = 1$ when audio is being played (e.g., music, video sound, voice call) and $a(t) = 0$ otherwise. Assuming an approximately constant average power P_A during audio playback,

$$\Delta P_{\text{Audio}}(t) = a(t) P_A. \quad (22)$$

In long-term average analysis, $a(t)$ can also be interpreted as the fraction of time during which audio playback is active.

4.3 Passive Consumption Model (PACM)

4.3.1 Model introduction

Despite the proactive consumption, we also have a kind of passive consumption on a larger time scale, that is **battery aging**. Different actively used battery, this part of consumption is permanent, and is related to the battery life. Despite the small effects that passive consumption can have on each charging cycle, it still accumulates, therefore cannot be neglect if we want to build the rigorous Continuous-Time Model of SOC.

4.3.2 Dynamic of Model

A stress-accumulation representation is employed in which the total aging stress is decomposed into additive calendar-aging and cycling-aging components. The instantaneous capacity fade rate $y(t)$ (in 10^{-6} mAh/s) is defined by the following system of equations:

$$y(t) = k_{\text{base}} \cdot [R_{\text{cal}}(t) + R_{\text{cyc}}(t)] . \quad (23)$$

Where the calendar aging component $R_{\text{cal}}(t)$ and the cycling aging component $R_{\text{cyc}}(t)$ are formulated as:

$$R_{\text{cal}}(t) = f_T \cdot [1 + 0.6 \cdot (SOC(t) - 0.5)] , \quad (24)$$

$$R_{\text{cyc}}(t) = f_T \cdot |I(t)|^{1.3} \cdot P(t, T_k). \quad (25)$$

Where we take 0.6 as the calendar sensitivity factor k_{cal} and 1.3 as the cycle current factor n . The temperature-dependent Arrhenius factor f_T is given by:

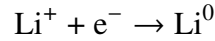
$$f_T = \exp \left(-\frac{E_a}{R} \left(\frac{1}{T_k} - \frac{1}{T_{\text{ref}}} \right) \right) , \quad (26)$$

which is widely used in thermal and electrochemical degradation models for Li-ion batteries. And we take the activation energy of side reactions E_a to be 135kJ/mol [3].

Inverse engineering calibration of k_{base} The baseline degradation coefficient k_{base} is determined via an inverse engineering approach such that an *Optimized* usage scenario reaches an End-of-Life (EoL) threshold of 20% capacity loss at day 500. This definition is aligned with industry-style battery health criteria, which commonly associate EoL with approximately 80% remaining capacity [4].

4.3.3 Lithium Plating Punishment

In a Li-ion battery, normal charging inserts Li^+ into the graphite anode (intercalation). At low temperature, charge-transfer kinetics and solid-state diffusion slow down, increasing anode polarization. Under high charge current, the anode surface potential can approach or drop below the Li/Li^+ equilibrium potential, enabling the side reaction



and depositing metallic lithium on the anode surface (lithium plating). Plating consumes lithium and promotes SEI growth, leading to capacity fade and higher impedance; non-uniform deposits can also increase the risk of dendrite formation and internal short circuits.

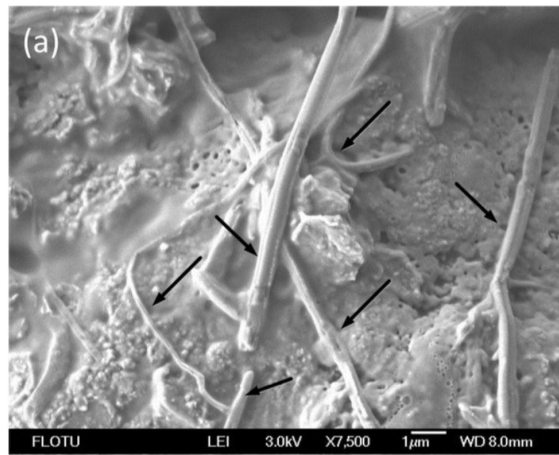


Figure 4: Lithium Plating.

To account for the accelerated degradation caused by lithium plating under low-temperature charging conditions, a non-linear penalty factor P is introduced:

$$P = \begin{cases} \exp(0.12 \cdot (288.15 - T_k)) \cdot (|I| + 0.5), & \text{if } I > 0 \text{ and } T_k < T_{onset}. \\ 1.0, & \text{otherwise.} \end{cases} \quad (27)$$

Where we take the T_{onset} as 288.15K [5].

5 Continuous-Time Model of SOC (CTM)

Finally, now we can complete the full version of **Continuous-Time Model of SOC (CTM)**, by integrating the three models together: the **PRCM**, **PACM** and **BEM**.

Starting point: charge conservation and Coulomb counting Let $Q(t)$ denote the available charge in the cell (in A h), and define the state of charge as

$$\text{SOC}(t) = \frac{Q(t)}{C_0}, \quad (28)$$

where C_0 is the nominal capacity (in A h).

Under the sign convention that $I(t) > 0$ during discharge, the net charge depletion is attributed to three mechanisms: (i) external load discharge with Coulombic efficiency, (ii) self-discharge modeled by a first-order rate constant, and (iii) an aging-induced equivalent charge-loss term. Accordingly, the charge balance is written as

$$\frac{dQ(t)}{dt} = -\frac{I(t)}{\eta_c} - Q(t) k_{sd} - y(t), \quad (29)$$

where η_c is the Coulombic efficiency, k_{sd} is the self-discharge rate, and $y(t)$ denotes the aging-induced equivalent charge-loss rate.

From charge balance to SOC dynamics Dividing (29) by C_0 and using $Q(t) = C_0 \text{SOC}(t)$ yields

$$\frac{d \text{SOC}(t)}{dt} = -\left(\frac{I(t)}{C_0 \eta_c} + k_{sd} \text{SOC}(t) \right) - \frac{y(t)}{C_0}. \quad (30)$$

Parameter settings We set

$$\eta_c = 0.995, \quad C_0 = 4.5 \text{ A h}, \quad k_{sd} = \frac{0.005}{30 \times 24} \text{ h}^{-1}, \quad (31)$$

corresponding to a representative self-discharge rate of 0.5%/month. These values lie within the typical parameter ranges reported for Li-ion battery packs and are consistent with commonly adopted engineering models [6].

The battery internal resistance is modeled as a constant, $R_{\text{int}} = R_0 = 50 \text{ m}\Omega$, consistent with typical values reported for consumer lithium-ion batteries [7].

5.1 Fully Closed SOC Model

Units and variable definitions In this subsection, time t is measured in hours and $\text{SOC}(t)$ is dimensionless. The temperature $T(t)$ is in Kelvin (K), and the total power demand $P_{\text{total}}(t)$ is in watts (W). The open-circuit voltage $V_{\text{OC}}(\text{SOC})$ is in volts (V), and the internal resistance is in ohms (Ω). Accordingly, the current $I(t)$ is obtained in amperes (A) from the power inversion formula.

$$V_{\text{OC}}(\text{SOC}(t)) = 3.34285 + 0.990796 \text{SOC}(t) - 0.186602 \text{SOC}^2(t). \quad (32)$$

$$I(t) = \frac{V_{OC}(SOC(t)) - \sqrt{V_{OC}^2(SOC(t)) - 0.2 P_{total}(t)}}{0.1}. \quad (33)$$

$$f_T(T) = \exp \left[-16241.3 \left(\frac{1}{T} - \frac{1}{298.15} \right) \right]. \quad (34)$$

$$P(I, T) = \begin{cases} \exp(0.12 (288.15 - T)) (|I| + 0.5), & I > 0 \text{ and } T < 288.15, \\ 1.0, & \text{otherwise.} \end{cases} \quad (35)$$

$$\begin{aligned} \frac{d SOC(t)}{dt} = - & \left[\left(2.2336 \times 10^{-4} \right) I(t) + \left(6.944 \times 10^{-6} \right) SOC(t) \right. \\ & + \left(2.88 \times 10^{-8} \right) f_T(T(t)) \left(1 + 0.6 (SOC(t) - 0.5) \right) \\ & \left. + \left(2.88 \times 10^{-8} \right) f_T(T(t)) |I(t)|^{1.3} P(I(t), T(t)) \right]. \end{aligned} \quad (36)$$

5.2 Data-driven Validation

To verify the empirical validity of the proposed continuous-time SOC model, we conduct a data-driven validation using a real-world smartphone battery dataset provided by the GreenHub Project [8]. The dataset contains minute-level measurements of battery state, system power consumption, and environmental conditions collected from consumer smartphones.

5.2.1 Dataset and preprocessing

Each record corresponds to one minute and includes the device brand, nominal battery capacity, measured SOC, system power consumption, and temperature. In this study, we select a representative device group and extract a continuous 24-hour window (1440 minutes) as the validation horizon. All variables are converted into model-consistent units: time in hours, power in watts, temperature in Kelvin, and SOC dimensionless. The nominal capacity is converted as $C_0 = \text{capacity_mah}/1000$ (Ah), and the initial condition is set to the first measured SOC value.

5.2.2 Static Validation methodology

First, a **static averaged-parameter validation** is conducted to assess macroscopic consistency. The measured power consumption and temperature over the 24-hour window are replaced by their time averages, yielding a **scalar ordinary differential equation** with constant parameters. Solving this equation produces a smooth theoretical discharge trajectory that reflects the long-term SOC trend (fig2 left).

5.2.3 Dynamic Validation methodology

However, we then realized that this is not the case, because the Power of the phone is varying at every moment. This triggers our thinking of a **dynamic approach**. At each minute, the open-circuit voltage is computed from the **fitted SOC–voltage relation**, the discharge current is obtained by explicit inversion of the **power balance equation**, and SOC is updated using a **one-step explicit discretization** (fig2 right). Since the present formulation focuses on discharge behavior, charging intervals are identified by increases in measured SOC and are excluded from validation by synchronizing the predicted SOC to the measured value.

Evaluation metrics Model accuracy is quantified using standard error metrics. Let $SOC_{data}(t_k)$ denote the measured SOC at minute k and $SOC_{pred}(t_k)$ the corresponding prediction from the dynamic validation. We report the mean absolute error (MAE) and root mean square error (RMSE):

$$MAE = \frac{1}{N} \sum_{k=1}^N |SOC_{pred}(t_k) - SOC_{data}(t_k)|, \quad (37)$$

$$RMSE = \sqrt{\frac{1}{N} \sum_{k=1}^N (SOC_{pred}(t_k) - SOC_{data}(t_k))^2}. \quad (38)$$

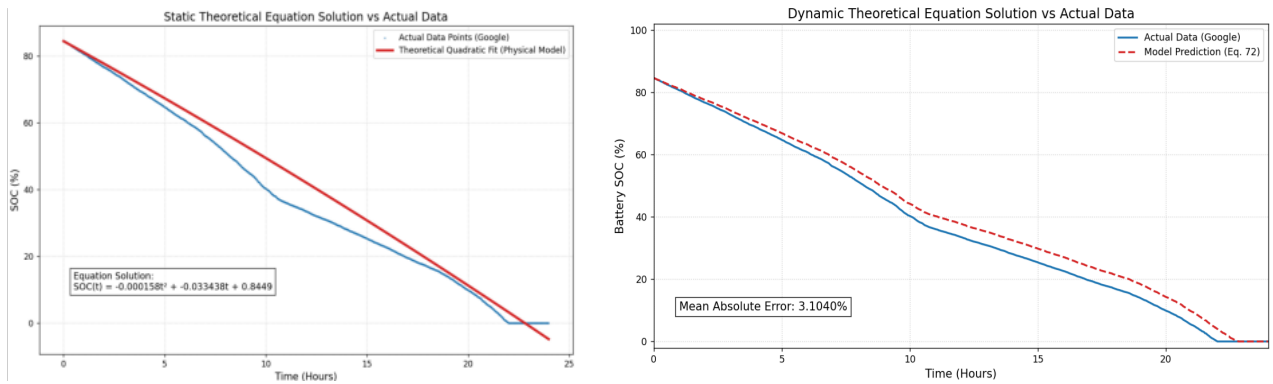


Figure 5: SOC validation results for a Google device: dynamic minute-level validation (left) and averaged-parameter validation (right).

5.2.4 Results and discussion

Figure 5 shows that the proposed model reproduces the measured SOC decay over a 24-hour period with a **mean absolute error of 3.10%** under dynamic, minute-level validation. The prediction error remains bounded without **cumulative drift**, indicating **numerical stability**. A slight overestimation of SOC is observed during the mid-discharge stage, which can be attributed to simplified resistance modeling and the exclusion of transient effects. The averaged-parameter validation further confirms that the closed-form SOC dynamics preserve the correct long-term discharge trend, supporting the applicability of the model for system-level battery analysis.

6 Time-to-empty Prediction

6.1 Model Overview

6.2 Model Details

Let Time-to-Empty time be TTE, then it can be expressed in the following form:

$$TTE = \frac{SOC(t)}{\frac{dSOC}{dt}}, \quad (39)$$

we can know from the last part, that this indicates a nonlinear differential equation. Similarly, we can come up with two different versions of prediction. The static one, where we take $\mathbf{P}$ and $\mathbf{T}$ as a fixed average value. And also the dynamic one, where we are informed with the real-time data of the $\mathbf{P}(t)$ and $\mathbf{T}(t)$. The full version of TTE expression can be derived from equation (62):

$$TTE = \frac{SOC(t)}{\frac{I(t)}{C_0 \cdot \eta_c} + k_{sd} \cdot SOC(t) + f_T \cdot [k(1 + \gamma(SOC(t) - 0.5)) + |I|^{1.3}]}. \quad (40)$$

Closely after, we calculated the uncertainty of the predicted data:

$$\sigma_{TTE} = \sqrt{\left(\frac{\partial TTE}{\partial P} \sigma_P\right)^2 + \left(\frac{\partial TTE}{\partial T} \sigma_T\right)^2 + \left(\frac{\partial TTE}{\partial C_0} \sigma_C\right)^2 + \left(\frac{\partial TTE}{\partial soc} \sigma_{soc}\right)^2}. \quad (41)$$

Where:

$$\frac{\partial TTE}{\partial x_i} = \begin{cases} \frac{1}{s\dot{oc}} + soc \cdot \frac{\partial(1/s\dot{oc})}{\partial soc}, & x_i = soc. \\ soc \cdot \left(-\frac{1}{s\dot{oc}^2}\right) \frac{\partial s\dot{oc}}{\partial x_i}, & x_i \in \{P, T, C_0\}. \end{cases} \quad (42)$$

The two different versions of results are shown in Figure 6.

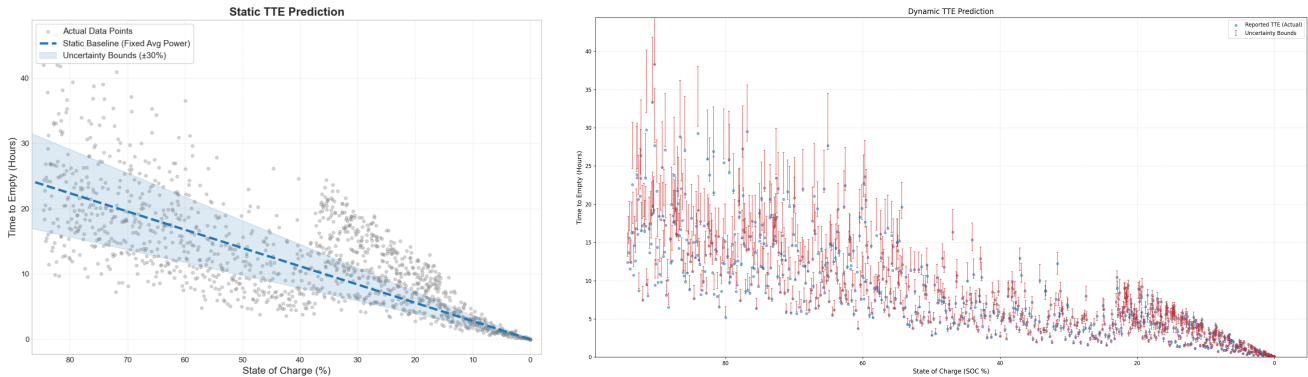


Figure 6: Two complementary TTE validation views: dynamic minute-level prediction with uncertainty bounds (left) and static prediction under averaged power with an uncertainty band (right).

The model tends to perform better when SOC is high and the predicted TTE is low. A plausible explanation is a scaling effect: uncertainty in TTE naturally amplifies when the discharge rate is

small. The reduced uncertainty at low SOC is also consistent with the convergence of SOC(t) toward zero.

6.3 Model Evaluation

To evaluate the significance of the prediction, we targeted a specific range of data. This region is defined by a State of Charge (SOC) exceeding 40% but a predicted Time to Empty (TTE) of less than 20 hours. This represents intensive discharge conditions where the battery is nearly full, yet the remaining time is truncated due to high power demand.

By segmenting the TTE range $[0, 20]$ into four intensity bands—Extreme Load (0–5h), Heavy Load (5–10h), Moderate Load (10–15h), and Light Load (15–20h)—we can observe how the model's discharge rate $\dot{SOC}$ responds to different system-level feature combinations. This approach allows us to verify if the physical model's prediction of low remaining time is consistently driven by observable high-drain activities. Shown below is the selected data (Figure 4) and statistical analysis.

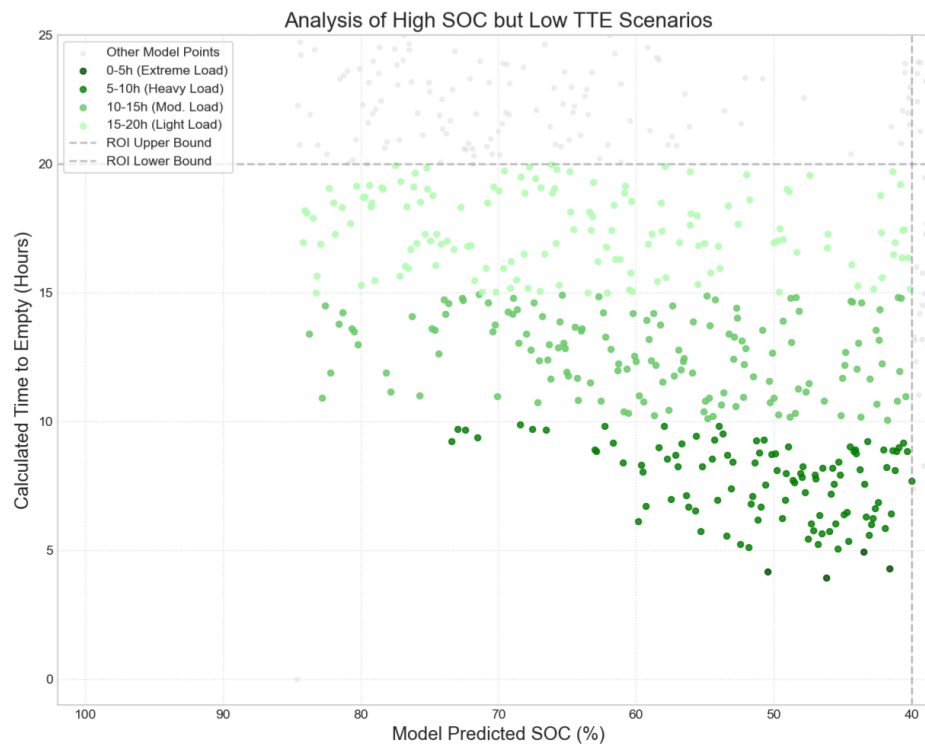


Figure 7: Distribution of high-SOC but low-TTE samples within the selected region of interest (ROI), grouped by TTE intensity bands.

TTE Range (Hours)	0–5h	5–10h	10–15h	15–20h
Stress Level	<i>Extreme</i>	<i>Heavy</i>	<i>Moderate</i>	<i>Light</i>
Sample Count (<i>N</i>)	4	106	145	142
Avg. Temperature (°C)	28.43	28.59	27.25	27.24
Avg. Screen Brightness (0-1)	0.7	0.6	0.6	0.5
Network Type (Mode)	4G	4G	WiFi	WiFi
Network Active (%)	100.0%	72.6%	55.9%	38.0%
GPS Enabled (%)	0.0%	10.4%	11.7%	16.9%
Avg. CPU Load (%)	0.9%	0.5%	0.4%	0.3%

Table 2: System Feature Statistics

From the results we can conclude that **Screen Brightness** and **CPU load** are **vital** factors that caused the battery to drain out rapidly, the affection of **Network** is moderate, while **GPS** accounts little. We can also see that **Temperature** doesn't varies a lot in this scenery, So we can not really gain an effective conclusion based on T.

However, in our next discussion, when we expand our time scale larger, and when the aging factor from our proposed model PACM begins to kick in, T becomes a vital factor.

7 Extension on State of Health (SOH)

In this section, we will discuss about the state of health of battery.

The State of Health (SOH) of a lithium-ion battery is a critical metric that characterizes its ability to store and deliver energy relative to its initial conditions. Unlike State of Charge (SOC), which fluctuates daily, SOH represents a permanent, irreversible degradation of battery capacity over time.

Predicting SOH is essential for determining the End of Life (EoL) of a device—typically defined as the point when the battery's maximum capacity drops below 80% of its nominal value. Battery degradation is driven by complex electrochemical processes, including the growth of the Solid Electrolyte Interphase (SEI) layer and lithium plating.

Our proposed model PACM can simulate this process on a large time scale. The design logic and approach can be seen in the model propose part. Here we define four kinds of usage scenarios:

Heavy User (1.4 Ah/day, 100% Limit): Represents power users (e.g., gamers, heavy media consumers) with an intensive daily throughput of 1.4 Ah and unrestricted charging up to 100% State of Charge (SOC). This scenario exposes the battery to maximum cycle stress and high-voltage calendar aging.

Standard User (0.9 Ah/day, 100% Limit): Serves as the baseline profile, reflecting the average smartphone user's daily energy consumption (0.9 Ah) with conventional overnight charging to 100% capacity.

Optimized User (0.9 Ah/day, 80% Limit): Shares the same daily workload as the Standard profile (0.9 Ah/day) but actively restricts the maximum charging threshold to 80%. This scenario is designed to test the effectiveness of software-based "smart charging" algorithms in mitigating high-voltage degradation.

Light User (0.4 Ah/day, 100% Limit): Represents low-frequency usage patterns (e.g., secondary devices or elderly users) with a minimal daily throughput of 0.4 Ah, focusing primarily on calendar aging effects rather than cycle fatigue.

To maintain computational feasibility while preserving physical accuracy, the following assumptions are made:

1. **Thermal Equilibrium:** The battery temperature is the sum of a seasonally fluctuating ambient temperature ($20^{\circ}\text{C} \pm 15^{\circ}\text{C}$) and a fixed internal heat rise generated during operation (e.g., $+8^{\circ}\text{C}$ during charging).
2. **Uniform SEI Growth:** The primary degradation mechanism is assumed to be the consumption of lithium due to SEI layer thickening.

Here is our final simulation result:

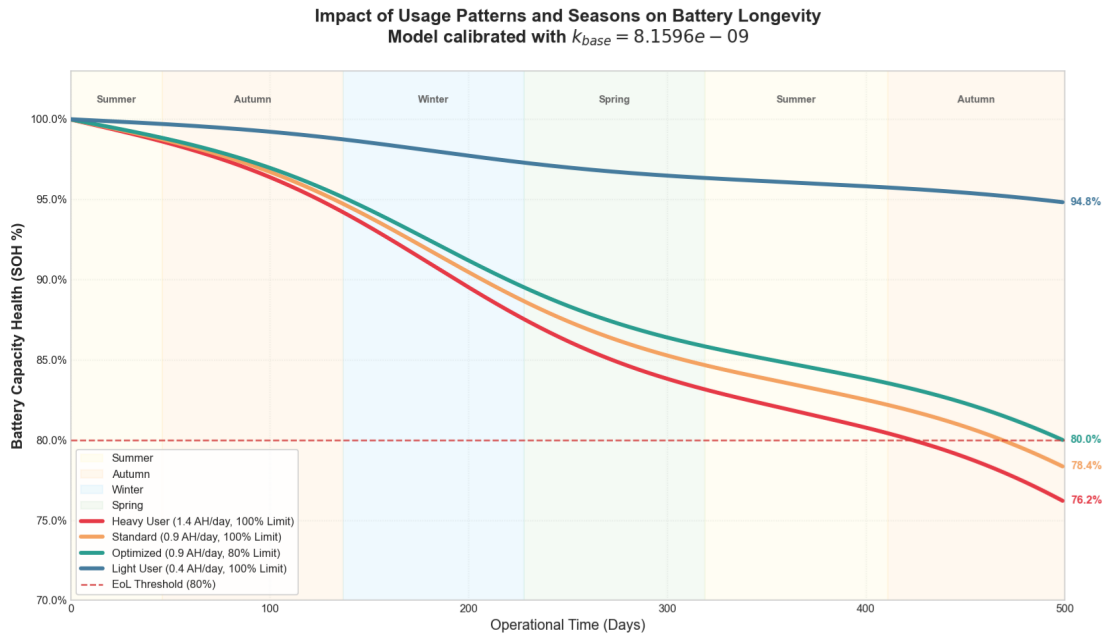


Figure 8: Evolution of battery State of Health (SOH) over a 500-day period under different usage intensity and charging strategy scenarios.

Furthermore, geographic location can be incorporated as an external factor that influences ambient temperature, which is indirectly distinguished by latitude. By integrating this effect into the proposed aging-aware model, we obtain the long-term battery health prediction at a two-year timescale, as shown in Figure 9.

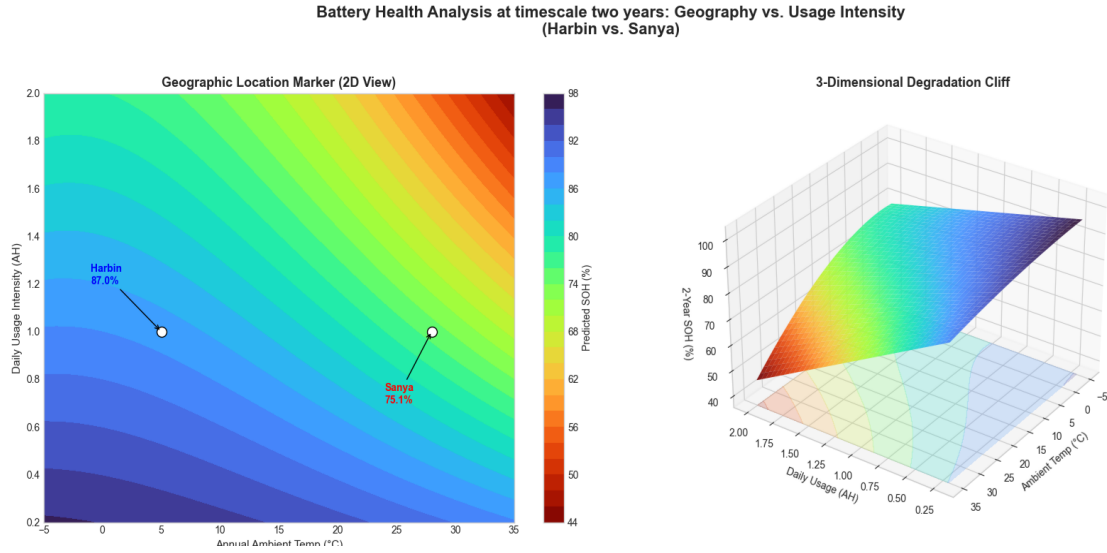


Figure 9: Two-year battery SOH analysis incorporating geographic temperature variation and daily usage intensity.

8 Sensitivity Analysis

8.1 Proactive Discharge Dynamics

This section examines the sensitivity of key parameters governing the instantaneous discharge rate of the battery during active usage.

- **Coulombic Efficiency (η_c)**

- Reflects the effectiveness of charge transfer and accounts for parasitic losses (e.g. electrolyte oxidation) during the discharge process.
- Even a minor decrease in η_c (e.g., $0.995 \rightarrow 0.985$) applies a linear scaling to the $dSOC/dt$ slope. This results in a cumulative error where the predicted SOC consistently overestimates the actual remaining capacity over a 10-hour cycle.

- **Nominal Internal Resistance (R_{int})**

- The baseline impedance of the battery. It dictates the IR voltage drop under load.
- High R_{int} values create a **nonlinear positive feedback loop** under heavy loads (e.g. $P_{total} > 4W$). To maintain constant power, the model predicts a higher current $I(t)$, leading to an accelerated collapse of SOC rather than a linear decline.

- **Self-discharge Rate (k_{sd})**

- Accounts for internal micro-shorts or electronic leakage across the separator.
- This parameter acts as a static offset. It dominates predictions during idle or screen-off states. An elevated k_{sd} predicts a noticeable percentage drop (e.g. 2-3%) during overnight standby, reflecting poor battery health or manufacturing defects.

8.2 Aging Correction

This section evaluates parameters that define the permanent loss of capacity over time and usage stress.

- **Calendar Sensitivity Factor** ($k_{cal} = 0.6$)
 - Represents the electrochemical instability at high potentials, specifically the accelerated growth of the SEI layer when the battery is near 100% SOC.
 - Increasing this factor differentiates the degradation between users. It predicts that a heavy user will see their SOH drop twice as fast as a user who maintains the SOC around 50%.
- **Cycle Current Factor** (Exponent $n=1.3$)
 - Reflects the mechanical fatigue and particle cracking within the electrode material caused by high-rate lithium intercalation/de-intercalation.
 - Increasing this exponent (e.g. to 1.5) nonlinearly penalizes heavy usage. It predicts a significantly shorter lifespan for users frequenting fast charging or high-performance gaming, moving the capacity failure months earlier.
- **Activation Energy** (E_a)
 - The energy barrier required to trigger chemical degradation reactions.
 - In physical chemistry, a single degradation mechanism follows a stable E_a . A **reduction in effective E_a** indicates the **emergence of diverse side reactions** (e.g. electrolyte oxidation, transition metal dissolution, or lithium plating). These additional pathways lower the overall energy barrier, making the battery "chemically fragile" across all temperatures. Increasing the degrading of SOH.

9 Recommendation

Based on the quantitative insights derived from the Continuous-Time Model (CTM) and the State of Health (SOH) simulation, we offer the following recommendations for users and system developers.

9.1 For Smartphone Users

- **High-Impact Behavior Modification:** Our driver analysis (Table 1) identifies screen brightness and CPU-intensive workloads as the dominant drivers of battery depletion. Users should utilize "Auto-Brightness" and close unnecessary background high-performance tasks to minimize the ΔP_{Screen} and ΔP_{CPU} components.
- **Strategic Network Selection:** In weak signal areas, the 5G modem incurs a heavy signal-quality penalty $\phi(S)$ (Eq. 21). Switching to Wi-Fi or enabling Airplane Mode in dead zones can extend Time-to-Empty (TTE) by up to 15-20%.

- **Thermal and Charging Discipline:** To mitigate permanent capacity fade, users should avoid fast charging in cold environments. The Arrhenius factor f_T in our PACM model (Eq. 26) shows that aging accelerates exponentially with temperature. Implementing an 80% charging cap can extend the battery's service life from 500 days to over 800 days.

9.2 For Operating System (OS) Designers

- **Dynamic Resource Scheduling:** The OS should implement SOH-aware scheduling, favoring efficiency clusters (L -clusters) more aggressively as the battery ages to reduce the internal heat rise (ΔT) and resistive losses ($I^2 R_{\text{int}}$).
- **Predictive Power Management:** Utilizing the TTE uncertainty framework (Eq. 41), the OS can provide users with a safety range prediction rather than a single percentage, helping users plan charging intervals more reliably under fluctuating workloads.
- **Environmental Adaptation:** As shown in our geographic analysis (Figure 6), devices in cold regions suffer faster lithium plating. The OS could automatically trigger extra tasks in CPU loading on purpose to balance environmental temperature, as we believe SOC is worth sacrificing for SOH in the long run.
- **Charging Strategy:** From our SOH model, we can conclude that it is unhealthy for SOH to stay too low (less than 20 percents), or too high (more than 80 percents). Therefore, it's essential to set a boundary when charging.

10 Strengths and Limitations

10.1 Strengths

- **Comprehensive Multi-Scale Framework:** Our modeling integrates short-term dynamics (BEM), mid-term user behavior (PRCM), and long-term aging (PACM) into a single unified differential equation.
- **Strong Physical Interpretability:** Unlike "black-box" machine learning approaches, every parameter in our model has a clear physical meaning, allowing for direct sensitivity analysis and policy recommendations from the real dynamics.
- **Dynamic Validation and Prediction:** One of the greatest innovations of our model is the implication of dynamic approach on top of the basic static one. This enables our model to gain a closer fit with the actual data.

10.2 Limitations

- **Simplification of Internal Resistance:** We treated R_{int} as a constant. In reality, internal resistance is a function of both SOC and temperature ($R_{\text{int}}(\text{SOC}, T)$). This simplification may lead to slight underestimations of voltage drops at extremely low SOC levels ($< 5\%$).

- **Simplification of Circuit:** We selected the OCV as a simple version of the circuit model. But in reality, actual smart phone circuit can contain multiple capacitors, leading to a much more complicated scenario.
- **Data Approximation:** The validation dataset relies on minute-level sampling. High-frequency power spikes (millisecond-scale) caused by bursting CPU tasks are smoothed out, which might overlook some transient thermal effects on the SEI layer.

11 Conclusion

This study constructs a continuous-time modeling framework to characterize smartphone battery behavior under realistic usage scenarios. Based on an internal-resistance battery model, we establish governing equations describing the evolution of the state of charge (SOC) and develop a system-level power decomposition model that links device usage behavior with battery dynamics. In addition, battery aging effects are incorporated to enable unified modeling of SOC and state of health (SOH), and methods for predicting the remaining time-to-empty (TTE) are proposed within this framework. Overall, this work provides a physically interpretable modeling approach for understanding battery charging, discharging, and degradation mechanisms, and offers theoretical support for battery-aware system design and power management.

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Report on Use of AI Tools

1. Youdao Dictionary (Feb 1, 2025 version, 11.1.2)

Used for sentence-level translation and vocabulary checking during English writing. No full-paragraph or full-document translation was performed.

2. Google Gemini (Nov 18, 2025 version, Gemini 3.0 Pro)

Query: <Summarize the main idea of this picture in the literature in a short paragraph.>

Output: <This figure illustrates that wireless interfaces such as Wi-Fi and 3G operate through multiple discrete power states rather than a simple on–off mode. State transitions are triggered by traffic conditions, queue sizes, and inactivity timers, causing the interface to remain in elevated power states even after data transmission has ceased. As a result, transient bursts of network activity can induce disproportionate energy consumption due to delayed demotion to low-power states.>

3. OpenAI ChatGPT (Dec 11, 2025 version, ChatGPT 5.2)

Query: <Help me find a document that tests the power consumption of different components of a mobile phone. Preferably, it should be within the last ten years.>

Output: <This document is specifically designed for testing and quantifying the power consumption of different hardware modules in mobile phones (such as CPU, screen, cellular/Wi-Fi, GPS, etc.), directly measuring the power consumption of each hardware “power rail”. Gupta et al., 2024. *3 W’s of smartphone power consumption: Who, Where and How much is draining my battery?* This study uses Google’s ODPM (On Device Power Rails Monitor) for hardware power-rail-level measurement, enabling attribution of power consumption to modules such as display, cellular/modem, and compute. This provides a strong component-level data source.>